Lec-11. 两个随机变量的函数

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两个随机变量函数的分布

- 已知随机变量 X, Y 的分布、二元函数 $g(x, y)$

$\Longrightarrow$ 求 $Z = g(X, Y)$ 的分布.

离散型随机变量函数的分布

若离散型随机变量 (X, Y) 联合分布律为

$$P\{X = x_i, Y = y_j\} = p_{ij},$$

则 $Z = g(X, Y)$ 的分布律为

$$\begin{aligned} P\{Z = z_k\} &= P\{g(X, Y) = z_k\} \\ &= \sum_{z_k = g(x_i, y_j)} p_{ij}, \qquad k = 1, 2, .. \end{aligned}$$

例

$X \backslash Y$	-2	-1	0
-1	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{3}{12}$
0	$\frac{2}{12}$	$\frac{1}{12}$	0
3	$\frac{2}{12}$	0	$\frac{2}{12}$

求 (1) $X+Y$, (2) $|X+Y|$ 的分布律.

解:

$X+Y$	-3	-2	-1	0	1	2	3
p_k	$\frac{1}{12}$	$\frac{3}{12}$	$\frac{4}{12}$	0	$\frac{2}{12}$	0	$\frac{2}{12}$

$\lvert X+Y\rvert$	1	2	3
p_k	$\frac{6}{12}$	$\frac{3}{12}$	$\frac{3}{12}$

$\square$

例

设两个独立的随机变量 X 与 Y 的分布律为

X	1	3
p_k	0.3	0.7

Y	2	4
p_k	0.6	0.4

求 $Z = X + Y$ 的分布律.

解.

X \ Y	2	4
1	0.18	0.12
3	0.42	0.28

$X+Y$	3	5	7
p_k	0.18	0.54	0.28

$\square$

例

X, Y 相互独立且具有同一分布律

X	0	1
p_k	$\frac{1}{2}$	$\frac{1}{2}$

求 $Z = \max\{X, Y\}$ 的分布律.

解:

X \ Y	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$

$\max\{X, Y\}$	0	1
p_k	$\frac{1}{4}$	$\frac{3}{4}$

$\square$

连续型随机变量函数的分布

已知连续型随机变量 (X, Y), 求 $Z = g(X, Y)$ 的分布函数或密度函数.

- 先求 Z 的分布函数,

$$\begin{aligned} F_Z(z) &= P\{Z \le z\} \\ &= P\{g(X, Y) \le z\} = \iint\limits_{g(x,y)\le z} f(x, y)\,dxdy. \end{aligned}$$

- 再通过求导得到概率密度函数,

$$f_Z(z) = F_Z'(z).$$

例

设 (X, Y) 的概率密度函数

$$f(x,y)=\begin{cases}3x & 0<y<x<1;\\ 0 & \text{其他}.\end{cases}$$

求 $Z=X-Y$ 的概率密度函数 $f_Z(z)$.

解:

$$
\begin{aligned}
F_Z(z) &= P\{Z \le z\} \\
&= P\{X - Y \le Z\} = \iint\limits_{x-y\le z} f(x,y)\,dxdy.
\end{aligned}
$$

z 的取值不同, 积分区域不同.

1. $z \le 0$ 时, 不与 $f(x,y)$ 的非零区域相交. $F_Z(z) = 0$.
2. $0 < z < 1$ 时,

$$
\begin{aligned}
\iint\limits_{x-y\le z} f(x,y)\,dxdy &= 1 - \iint\limits_{x-y>z} f(x,y)\,dxdy \\
&= 1 - \int_z^1 \int_0^{x-z} 3x\,dydx = \frac{3}{2}z - \frac{1}{2}z^3.
\end{aligned}
$$

3. $z \ge 1$ 时, $F_Z(z) = 1$.

故 $f_Z(z) = \begin{cases} \frac{3(1-z^2)}{2} & 0 < z < 1; \\ 0 & \text{其他}. \end{cases}$ □

连续型随机变量的三种函数

若 X, Y 为连续型随机变量，则

- $Z = X + Y$;
- $Z = XY, Z = \frac{Y}{X}$;
- $Z = \max\{X, Y\}, Z = \min\{X, Y\}$;

仍为连续型的随机变量.

$Z = X + Y$ 的分布

$$F_Z(z) = P\{Z \leq z\} = P\{X + Y \leq z\}$$

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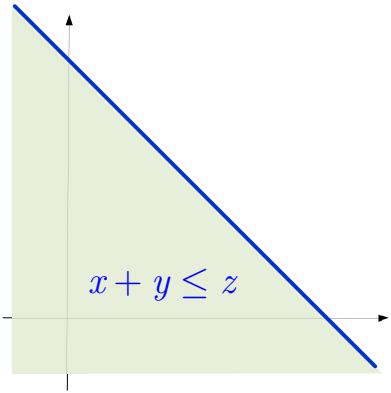
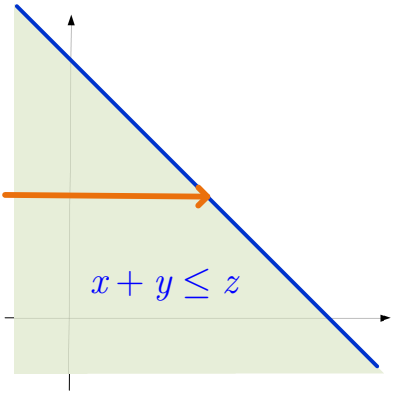


$Z = X + Y$ 的分布

$$F_Z(z) = P\{Z \le z\} = P\{X + Y \le z\}$$

$$= \iint\limits_{x+y\le z} f(x, y)\,dxdy$$

$$\overset{\text{化累次积分}}{=\!=\!=} \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{z-y} f(x, y)\,dx \right) dy$$



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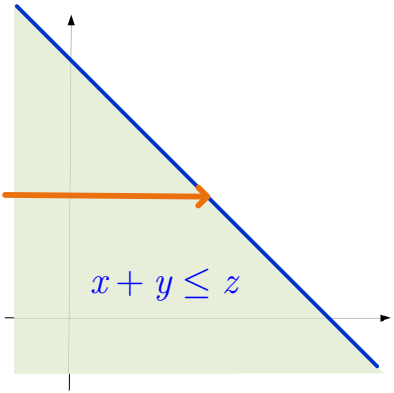
&\overset{u=x+y}{=\!=\!=} \int_{-\infty}^{+\infty}\left(\int_{-\infty}^{z} f(u-y,y)\,du\right)dy
\end{aligned}
$$



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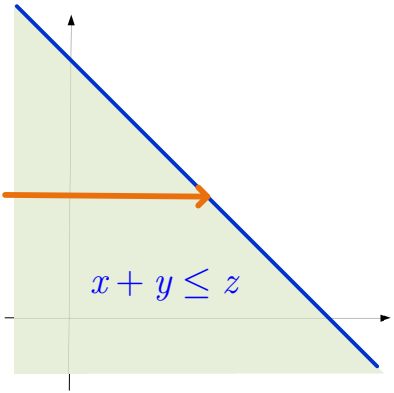

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F_Z(z) &= P\{Z \le z\} = P\{X + Y \le z\} \\
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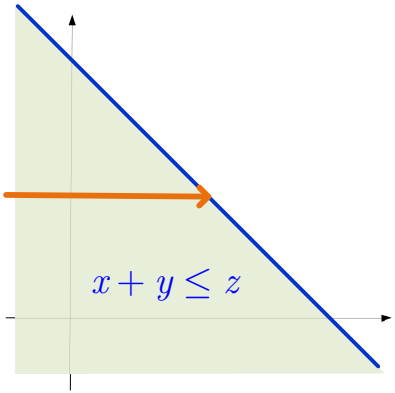
&\overset{\text{化累次积分}}{=\!=\!=} \int_{-\infty}^{+\infty}\left(\int_{-\infty}^{z-y} f(x,y)\,dx\right)dy \\
&\overset{u=x+y}{=\!=\!=} \int_{-\infty}^{+\infty}\left(\int_{-\infty}^{z} f(u-y,y)\,du\right)dy \\
&= \int_{-\infty}^{z}\left(\int_{-\infty}^{+\infty} f(u-y,y)\,dy\right)du \triangleq \int_{-\infty}^{z} f_Z(u)\,du
\end{aligned}
$$



$Z = X + Y$ 的分布

- 所以 $Z = X + Y$ 的概率密度为

$$f_Z(z) = \int_{-\infty}^{+\infty} f(z - y, y)\, dy.$$

$Z = X + Y$ 的分布

- 所以 $Z = X + Y$ 的概率密度为

$$f_Z(z) = \int_{-\infty}^{+\infty} f(z - y, y)\, dy.$$

- 由对称性,

$$f_Z(z) = \int_{-\infty}^{+\infty} f(x, z - x)\, dx.$$

- 若 X 和 Y 相互独立, 则

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y)\, dy$$
$$= \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\, dx.$$

- 上面公式称为函数 f_X 和 $f_Y(y)$ 的卷积公式，记为 $f_X * f_Y$. 即

$$f_X * f_Y = \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y)\, dy$$
$$= \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\, dx.$$

例

设 X 和 Y 是相互独立的, 且都服从 $N(0,1)$. 求 $Z=X+Y$ 的概率密度函数.

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设 X 和 Y 是相互独立的, 且都服从 $N(0,1)$. 求 $Z=X+Y$ 的概率密度函数.

解:

$$
\begin{aligned}
f_Z(z) &= \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\,dx = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} \cdot e^{-\frac{(z-x)^2}{2}}\,dx \\
&= \frac{1}{2\pi} e^{-\frac{z^2}{4}} \int_{-\infty}^{+\infty} e^{-(x-\frac{z}{2})^2}\,dx = \frac{1}{2\pi} e^{-\frac{z^2}{4}} \int_{-\infty}^{+\infty} e^{-t^2}\,dt \\
&= \frac{1}{2\pi} e^{-\frac{z^2}{4}} \cdot \sqrt{\pi} = \frac{1}{2\sqrt{\pi}} e^{-\frac{z^2}{4}}.
\end{aligned}
$$

所以 $Z \sim N(0,2)$. $\square$

一般情况

性质

- 若 X, Y 相互独立且 $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, 则 $Z = X + Y$ 仍然服从正态分布且

$$Z \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

- n 个独立正态随机变量的线性组合仍服从正态分布. 即设 $X_i \sim N(\mu_i, \sigma_i^2)$ 且相互独立, 则

$$c_0 + c_1 X_1 + \cdots + c_n X_n \sim N(\mu, \sigma^2),$$

其中 $c_0, c_1, \cdots, c_n$ 是不全为 0 的常数,
$\mu = c_0 + c_1\mu_1 + \cdots + c_n\mu_n$, $\sigma^2 = c_1^2\sigma_1^2 + \cdots + c_n^2\sigma_n^2$.

例

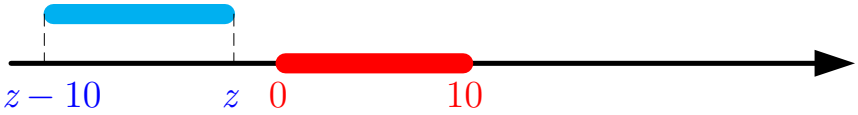
在一简单电路中, 两电阻 R_1 和 R_2 串联, 设 R_1, R_2 相互独立, 它们的概率密度均为

$$f(x) = \begin{cases} \frac{10-x}{50} & 0 \le x \le 10; \\ 0 & \text{其他}. \end{cases}$$

求总电阻 $R = R_1 + R_2$ 的概率密度.

解: $f_R(z) = \int_{-\infty}^{+\infty} f_{R_1}(x) f_{R_2}(z-x)\,dx,$

被积函数不为 0$\Leftrightarrow \begin{cases} 0 < x < 10; \\ 0 < z - x < 10 \end{cases} \Rightarrow \begin{cases} 0 < x < 10; \\ z - 10 < x < z \end{cases}$



$$f_R(z) = \begin{cases} 0 & z < 0; \\ & \end{cases}$$

解: $f_R(z) = \int_{-\infty}^{+\infty} f_{R_1}(x) f_{R_2}(z-x)\,dx,$

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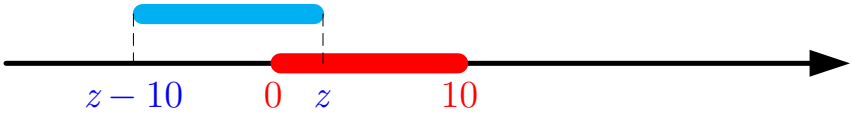

$$f_R(z) = \begin{cases} 0 & z < 0; \\ \int_0^z \frac{10-x}{50} \times \frac{10-(z-x)}{50}\,dx & 0 \le z < 10; \end{cases}$$

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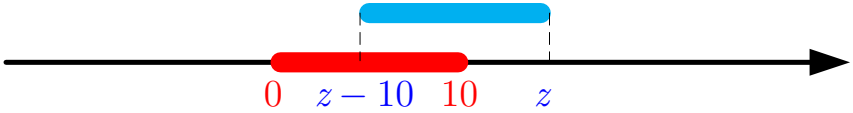
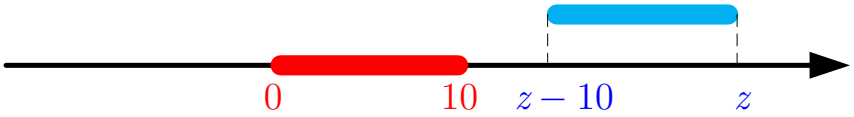


$$f_R(z) = \begin{cases} 0 & z < 0; \\ \int_0^z \frac{10-x}{50} \times \frac{10-(z-x)}{50}\,dx & 0 \le z < 10; \\ \int_{z-10}^{10} \frac{10-x}{50} \times \frac{10-(z-x)}{50}\,dx & 10 \le z < 20; \end{cases}$$

解: $f_R(z) = \int_{-\infty}^{+\infty} f_{R_1}(x) f_{R_2}(z-x)\,dx,$

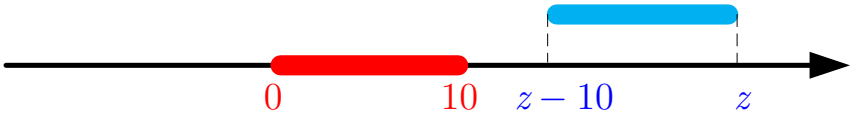
被积函数不为 0$\Leftrightarrow \begin{cases} 0 < x < 10; \\ 0 < z - x < 10 \end{cases} \Rightarrow \begin{cases} 0 < x < 10; \\ z - 10 < x < z \end{cases}$



$$f_R(z) = \begin{cases} 0 & z < 0; \\ \int_0^z \frac{10-x}{50} \times \frac{10-(z-x)}{50}\,dx & 0 \le z < 10; \\ \int_{z-10}^{10} \frac{10-x}{50} \times \frac{10-(z-x)}{50}\,dx & 10 \le z < 20; \\ 0 & 20 \le z \end{cases}$$

解: $f_R(z) = \int_{-\infty}^{+\infty} f_{R_1}(x) f_{R_2}(z-x)\,dx,$

被积函数不为 0$\Leftrightarrow \begin{cases} 0 < x < 10; \\ 0 < z - x < 10 \end{cases} \Rightarrow \begin{cases} 0 < x < 10; \\ z - 10 < x < z \end{cases}$



$$f_R(z) = \begin{cases} 0 & z < 0; \\ \int_0^z \frac{10-x}{50} \times \frac{10-(z-x)}{50}\,dx & 0 \le z < 10; \\ \int_{z-10}^{10} \frac{10-x}{50} \times \frac{10-(z-x)}{50}\,dx & 10 \le z < 20; \\ 0 & 20 \le z \end{cases}$$

$$= \begin{cases} \frac{1}{15000}(600z - 60z^2 + z^3) & 0 \le z < 10; \\ \frac{1}{15000}(20-z)^3 & 10 \le z < 20; \\ 0 & \text{其他}. \end{cases}$$

例

设 X, Y 相互独立, 且分别服从参数为 α, θ; β, θ 的 Γ 分布 ($X \sim \Gamma(\alpha, \theta)$, $Y \sim \Gamma(\beta, \theta), \alpha, \beta, \theta > 0$), 概率密度分别为

$$f_X(x) = \begin{cases} \frac{1}{\theta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\theta}} & x > 0; \\ 0 & \text{其他}. \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{\theta^\beta \Gamma(\beta)} y^{\beta-1} e^{-\frac{y}{\theta}} & y > 0; \\ 0 & \text{其他}. \end{cases}$$

证 $Z = X + Y$ 服从参数为 $\alpha + \beta, \theta$ 的 Γ 分布, 即 $X + Y \sim \Gamma(\alpha + \beta, \theta)$.

Γ 函数和 B 函数

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx, \qquad \alpha > 0$$

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1} dt, \qquad \alpha, \beta > 0$$

Γ 函数和 B 函数

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等式关系:

- $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$, $\Gamma(n+1) = \int_0^\infty x^n e^{-x} dx = n!$;
- $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.

证明: $Z = X + Y$ 的概率密度为 $f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\,dx$,

被积函数不为 0 时 $\Leftrightarrow \begin{cases} x > 0; \\ z - x > 0 \end{cases}$

(1) $z < 0$, $f_Z(z) = 0$,

(2) $z > 0$,

$$
\begin{aligned}
f_Z(z) &= \int_0^z \frac{1}{\theta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\theta}} \frac{1}{\theta^\beta \Gamma(\beta)} (z-x)^{\beta-1} e^{-\frac{(z-x)}{\theta}}\,dx \\
&= \frac{e^{-\frac{z}{\theta}}}{\theta^{\alpha+\beta}\Gamma(\alpha)\Gamma(\beta)} \int_0^z x^{\alpha-1}(z-x)^{\beta-1}\,dx \\
&\overset{x=zt}{=\!=\!=} \frac{z^{\alpha+\beta-1} e^{-\frac{z}{\theta}}}{\theta^{\alpha+\beta}\Gamma(\alpha)\Gamma(\beta)} \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,dt \\
&= \frac{B(\alpha,\beta)}{\theta^{\alpha+\beta}\Gamma(\alpha)\Gamma(\beta)} z^{\alpha+\beta-1} e^{-\frac{z}{\theta}} = \frac{1}{\theta^{\alpha+\beta}\Gamma(\alpha+\beta)} z^{\alpha+\beta-1} e^{-\frac{z}{\theta}}.
\end{aligned}
$$

所以

$$f_Z(z) = \begin{cases} \frac{1}{\theta^{\alpha+\beta}\Gamma(\alpha+\beta)} z^{\alpha+\beta-1} e^{-\frac{z}{\theta}} & z > 0; \\ 0 & \text{其他}. \end{cases}$$

即 $X + Y \sim \Gamma(\alpha+\beta, \theta)$. $\square$

所以

$$f_Z(z) = \begin{cases} \frac{1}{\theta^{\alpha+\beta}\Gamma(\alpha+\beta)} z^{\alpha+\beta-1} e^{-\frac{z}{\theta}} & z > 0; \\ 0 & \text{其他}. \end{cases}$$

即 $X + Y \sim \Gamma(\alpha+\beta, \theta)$. □

性质 (Γ 分布可加性)

若 $X_1, \cdots, X_n$ 相互独立且 $X_i \sim \Gamma(\alpha_i, \beta)$. 则

$$X_1 + \cdots + X_n \sim \Gamma(\alpha_1 + \cdots + \alpha_n, \beta).$$

$Z=\frac{Y}{X}$ 和 $Z=XY$ 的分布

设 (X, Y) 为连续型随机变量, 则 $Z=\frac{Y}{X}$, $Z=XY$ 的概率密度为

$$f_{Y/X}(z)=\int_{-\infty}^{+\infty}|x| f(x, xz) dx,$$

$$f_{XY}(z)=\int_{-\infty}^{+\infty}\frac{1}{|x|} f\left(x, \frac{z}{x}\right) dx.$$

$$F_{Y/X}(z) = P\{\, Y/X \leq z\} = \iint\limits_{y/x \leq z} f(x, y)\, dx dy$$

$$
\begin{aligned}
F_{Y/X}(z) = P\{\, Y/X \leq z\} &= \iint\limits_{y/x \leq z} f(x, y)\, dxdy \\
&= \iint\limits_{y/x \leq z, x<0} f(x, y)\, dxdy + \iint\limits_{y/x \leq z, x>0} f(x, y)\, dxdy \\
&= \int_{-\infty}^{0} \int_{zx}^{+\infty} f(x, y)\, dydx + \int_{0}^{+\infty} \int_{-\infty}^{zx} f(x, y)\, dydx
\end{aligned}
$$

$$
\begin{aligned}
F_{Y/X}(z) = P\{\, Y/X \leq z\} &= \iint\limits_{y/x \leq z} f(x, y)\, dxdy \\
&= \iint\limits_{y/x \leq z, x<0} f(x, y)\, dxdy + \iint\limits_{y/x \leq z, x>0} f(x, y)\, dxdy \\
&= \int_{-\infty}^{0} \int_{zx}^{+\infty} f(x, y)\, dydx + \int_{0}^{+\infty} \int_{-\infty}^{zx} f(x, y)\, dydx \\
&\overset{y=xu}{=\!=\!=} \int_{-\infty}^{0} \int_{z}^{-\infty} xf(x, xu)\, dudx + \int_{0}^{+\infty} \int_{-\infty}^{z} xf(x, xu)\, dudx
\end{aligned}
$$

$$
\begin{aligned}
F_{Y/X}(z) = P\{Y/X \le z\} &= \iint\limits_{y/x \le z} f(x,y)\,dxdy \\
&= \iint\limits_{y/x \le z, x<0} f(x,y)\,dxdy + \iint\limits_{y/x \le z, x>0} f(x,y)\,dxdy \\
&= \int_{-\infty}^{0}\int_{zx}^{+\infty} f(x,y)\,dydx + \int_{0}^{+\infty}\int_{-\infty}^{zx} f(x,y)\,dydx \\
&\overset{y=xu}{=\!=\!=} \int_{-\infty}^{0}\int_{z}^{-\infty} xf(x,xu)\,dudx + \int_{0}^{+\infty}\int_{-\infty}^{z} xf(x,xu)\,dudx \\
&= \int_{-\infty}^{0}\int_{-\infty}^{z} (-x)f(x,xu)\,dudx + \int_{0}^{+\infty}\int_{-\infty}^{z} xf(x,xu)\,dudx
\end{aligned}
$$

$$
\begin{aligned}
F_{Y/X}(z) = P\{\, Y/X \le z\} &= \iint\limits_{y/x \le z} f(x,y)\,dxdy \\
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&\overset{y=xu}{=\!=\!=} \int_{-\infty}^{0}\int_{z}^{-\infty} xf(x,xu)\,dudx + \int_{0}^{+\infty}\int_{-\infty}^{z} xf(x,xu)\,dudx \\
&= \int_{-\infty}^{0}\int_{-\infty}^{z} (-x)f(x,xu)\,dudx + \int_{0}^{+\infty}\int_{-\infty}^{z} xf(x,xu)\,dudx \\
&= \int_{-\infty}^{+\infty}\int_{-\infty}^{z} |x|f(x,xu)\,dudx = \int_{-\infty}^{z}\left(\int_{-\infty}^{+\infty} |x|f(x,xu)\,dx\right) du
\end{aligned}
$$

- 类似可证 $Z = XY$ 的概念密度 (作业)

$$f_{XY}(z) = \int_{-\infty}^{+\infty} \frac{1}{|x|} f\left(x, \frac{z}{x}\right) dx.$$

- 特别地, X, Y 相互独立时,

$$f_{Y/X}(z) = \int_{-\infty}^{+\infty} |x| f_X(x) f_Y(zx) dx,$$

$$f_{XY}(z) = \int_{-\infty}^{+\infty} \frac{1}{|x|} f_X(x) f_Y\left(\frac{z}{x}\right) dx.$$

例

某公司提供一种地震保险. 保费 Y 的概率密度为

$$f(y)=\begin{cases}\frac{y}{25}e^{-y/5} & y>0;\\ 0 & \text{其他}.\end{cases}$$

保险赔付 X 的概率密度为

$$f(x)=\begin{cases}\frac{1}{5}e^{-x/5} & x>0;\\ 0 & \text{其他}.\end{cases}$$

设 X 与 Y 相互独立, 求 $Z=Y/X$ 的概率密度.

解: 当 $z<0$ 时, $f_Z(z)=0$.

当 $z>0$ 时,

$$
\begin{aligned}
f_Z(z) &= \int_{-\infty}^{+\infty} |\, x \,| f_X(x) f_Y(zx)\, dx \\
&= \int_0^{+\infty} x \cdot \frac{1}{5} e^{-x/5} \cdot \frac{xz}{25} e^{-xz/5}\, dx \\
&= \frac{z}{125} \int_0^{+\infty} x^2 e^{-x \cdot \frac{1+z}{5}}\, dx \\
&\overset{t=x(1+z)/5}{=\!=\!=\!=} \frac{z}{125((1+z)/5)^3} \int_0^{+\infty} t^2 e^{-t}\, dt \\
&= \frac{z \cdot 2!}{(1+z)^3} = \frac{2z}{(1+z)^3}.
\end{aligned}
$$

$\square$

$M = \max\{X, Y\}$ 和 $N = \min\{X, Y\}$ 的分布

设 X, Y 是相互独立的随机变量,

- $$\begin{aligned} F_{\max}(z) &= P\{M \le z\} = P\{X \le z, Y \le z\} \\ &= P\{X \le z\}P\{Y \le z\} = F_X(z)F_Y(z) \end{aligned}$$

$M = \max\{X, Y\}$ 和 $N = \min\{X, Y\}$ 的分布

设 X, Y 是相互独立的随机变量,

- $$\begin{aligned} F_{\max}(z) &= P\{M \le z\} = P\{X \le z, Y \le z\} \\ &= P\{X \le z\}P\{Y \le z\} = F_X(z)F_Y(z) \end{aligned}$$
- $$\begin{aligned} F_{\min}(z) &= P\{N \le z\} = 1 - P\{N > z\} \\ &= 1 - P\{X > z, Y > z\} \\ &= 1 - P\{X > z\}P\{Y > z\} \\ &= 1 - [1 - F_X(z)][1 - F_Y(z)] \end{aligned}$$

$M = \max\{X_1, \cdots, X_n\}$ 和 $N = \min\{X_1, \cdots, X_n\}$ 的分布

设 $X_1, \cdots, X_n$ 是 n 个相互独立的随机变量，

- $F_{\max}(z) = F_{X_1}(z) \cdots F_{X_n}(z)$,

$M = \max\{X_1, \cdots, X_n\}$ 和 $N = \min\{X_1, \cdots, X_n\}$ 的分布

设 $X_1, \cdots, X_n$ 是 n 个相互独立的随机变量，

- $F_{\max}(z) = F_{X_1}(z) \cdots F_{X_n}(z)$,
- $F_{\min}(z) = 1 - [1 - F_{X_1}(z)] \cdots [1 - F_{X_n}(z)]$.

$M = \max\{X_1, \cdots, X_n\}$ 和 $N = \min\{X_1, \cdots, X_n\}$ 的分布

设 $X_1, \cdots, X_n$ 是 n 个相互独立的随机变量,

- $F_{\max}(z) = F_{X_1}(z) \cdots F_{X_n}(z)$,
- $F_{\min}(z) = 1 - [1 - F_{X_1}(z)] \cdots [1 - F_{X_n}(z)]$.
- 特别地, 当 $X_1, \cdots, X_n$ 有相同分布函数时

$$F_{\max}(z) = [F(z)]^n, \quad F_{\min} = 1 - [1 - F(z)]^n.$$

例

设系统 L 由两个相互独立的子系统 L_1, L_2 连接而成, 连接的方式分别为 (i) 串联, (ii) 并联, (iii) 备用 (当 L_1 损坏时, L_2 开始工作)

设 L_1, L_2 的寿命分别为 X, Y. 已知其概率密度为

$$f_X(x) = \begin{cases} \alpha e^{-\alpha x} & x > 0; \\ 0 & \text{其他}. \end{cases}$$

$$f_Y(y) = \begin{cases} \beta e^{-\beta y} & y > 0; \\ 0 & \text{其他}. \end{cases}$$

其中 $\alpha > 0$, $\beta > 0$, 且 $\alpha \neq \beta$, 试分别就三种连接方式写出 L 的寿命 Z 的概率密度.

解:(i) 串联, 由于 L_1, L_2 中一个损坏时, 系统 L 就停止工作, 则 L 的寿命为 $Z = \min\{X, Y\}$.

$$F_X(x) = \begin{cases} 1 - e^{-\alpha x} & x > 0; \\ 0 & \text{其他}. \end{cases}$$

$$F_Y(y) = \begin{cases} 1 - e^{-\beta y} & y > 0; \\ 0 & \text{其他}. \end{cases}$$

$$\begin{aligned} F_{\min}(z) &= \begin{cases} 1 - (1 - F_X(z))(1 - F_Y(z)) & z > 0; \\ 0 & \text{其他}. \end{cases} \\ &= \begin{cases} 1 - e^{-(\alpha+\beta)z} & z > 0; \\ 0 & \text{其他}. \end{cases} \end{aligned}$$

$$f_{\min}(z) = \begin{cases} (\alpha + \beta) e^{-(\alpha+\beta)z} & z > 0; \\ 0 & \text{其他}. \end{cases}$$

(ii) 并联, $Z = \max\{X, Y\}$

$$F_{\max}(z) = F_X(z)F_Y(z) = \begin{cases} (1-e^{-\alpha z})(1-e^{-\beta z}) & z > 0; \\ 0 & \text{其他}. \end{cases}$$

$$f_{\max}(z) = \begin{cases} \alpha e^{-\alpha z} + \beta e^{\beta z} - (\alpha+\beta)e^{-(\alpha+\beta)z} & z > 0; \\ 0 & \text{其他}. \end{cases}$$

(iii) 备用情况 $Z = X + Y$.

$$
\begin{aligned}
f(z) &= \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y)\, dy \\
&= \int_0^z \alpha e^{-\alpha(z-y)} \beta e^{-\beta y} dy \\
&= \alpha\beta e^{-\alpha z} \int_0^z e^{-(\beta-\alpha)y} dy \\
&= \frac{\alpha\beta}{\beta-\alpha}(e^{-\alpha z} - e^{-\beta z}), \qquad z > 0.
\end{aligned}
$$

$$
f(z) = \begin{cases} \frac{\alpha\beta}{\beta-\alpha}(e^{-\alpha z} - e^{-\beta z}) & z > 0; \\ 0 & \text{其他}. \end{cases}
$$

□

$$
\begin{aligned}
f_{X+Y}(z) &= \int_{-\infty}^{+\infty} f(x, z-x)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\,dx \\
&= \int_{-\infty}^{+\infty} f(z-y, y)\,dy \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y)\,dy.
\end{aligned}
$$

$$f_{X+Y}(z) = \int_{-\infty}^{+\infty} f(x, z-x)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\,dx$$

$$= \int_{-\infty}^{+\infty} f(z-y, y)\,dy \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y)\,dy.$$

$$f_{Y/X}(z) = \int_{-\infty}^{+\infty} |\,x\,|\, f(x, xz)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} |\,x\,|\, f_X(x) f_Y(xz)\,dx$$

$$f_{XY}(z) = \int_{-\infty}^{+\infty} \frac{1}{|\,x\,|} f\left(x, \frac{z}{x}\right)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} \frac{1}{|\,x\,|} f_X(x) f_Y\left(\frac{z}{x}\right)\,dx.$$

$$f_{X+Y}(z) = \int_{-\infty}^{+\infty} f(x, z-x)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\,dx$$

$$= \int_{-\infty}^{+\infty} f(z-y, y)\,dy \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y)\,dy.$$

$$f_{Y/X}(z) = \int_{-\infty}^{+\infty} |\,x\,|\, f(x, xz)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} |\,x\,|\, f_X(x) f_Y(xz)\,dx$$

$$f_{XY}(z) = \int_{-\infty}^{+\infty} \frac{1}{|\,x\,|} f\left(x, \frac{z}{x}\right)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} \frac{1}{|\,x\,|} f_X(x) f_Y\left(\frac{z}{x}\right)\,dx.$$

$$F_{max}(z) = F(z, z) \xlongequal{X,Y\text{独立}} F_X(z) F_Y(z).$$

$$f_{X+Y}(z) = \int_{-\infty}^{+\infty} f(x, z-x)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x)\,dx$$

$$= \int_{-\infty}^{+\infty} f(z-y, y)\,dy \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y)\,dy.$$

$$f_{Y/X}(z) = \int_{-\infty}^{+\infty} |\,x\,|\, f(x, xz)\,dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} |\,x\,|\, f_X(x) f_Y(xz)\,dx$$

$$f_{XY}(z) = \int_{-\infty}^{+\infty} \frac{1}{|\,x\,|} f\left(x, \frac{z}{x}\right) dx \xlongequal{X,Y\text{独立}} \int_{-\infty}^{+\infty} \frac{1}{|\,x\,|} f_X(x) f_Y\left(\frac{z}{x}\right) dx.$$

$$F_{max}(z) = F(z, z) \xlongequal{X,Y\text{独立}} F_X(z) F_Y(z).$$

$$F_{min}(z) = F_X(z) + F_Y(z) - F(z, z)$$

$$\xlongequal{X,Y\text{独立}} F_X(z) + F_Y(z) - F_X(z) F_Y(z)$$

$$\xlongequal{X,Y\text{独立}} 1 - [1 - F_X(z)][1 - F_Y(z)].$$